

Difference between python2 and python3

Key Syntax Changes

- **Print:** Python 2 uses a statement, Python 3 uses a function:

```
# Python 2
print "Hello"
```

```
# Python 3
print("Hello")
```

- **Division:** In Python 2, `/` between integers truncates to int; Python 3 returns a float. Use `//` for integer division in Python 3.

```
# Python 2: 5 / 2 -> 2
# Python 3: 5 / 2 -> 2.5
```

- **Exceptions:** Python 3 requires `as` a keyword.

```
# Python 2
except ValueError, e:
```

```
# Python 3
except ValueError as e:
```

What is PEP 8 and Why is it Important in Python?

PEP 8, or the Python Enhancement Proposal 8, is the official style guide for Python code. It was created by Guido van Rossum, Barry Warsaw, and Nick Coghlan in 2001 to ensure that Python code remains readable and consistent, regardless of the developer writing it. PEP 8 is part of a broader collection of Python Enhancement Proposals (PEPs), which are design documents providing guidelines and new features for Python. PEP 8 was introduced to standardize the formatting of Python code, as Python grew in popularity, and different developers had different coding styles, leading to confusion and making it difficult to read and maintain code written by others. PEP 8 addresses this issue by providing a comprehensive set of

conventions for writing Python code, covering everything from indentation and line length to naming conventions and comment practices. The main purpose of PEP 8 is to improve the readability of Python code. By following a common set of rules, developers can ensure that their code is easy to understand and that others can easily collaborate on projects. PEP 8 is not just about aesthetics; it's about writing code that is clean, predictable, and consistent.